

Work Flow

1 Team Lead (Tum) – Architecture + Risk Engine Owner

Tumhara kaam sabse critical hai.

Responsibilities:

- Final risk definition freeze karna (jab decide ho)
- Risk formula implement karna
- Seasonal analysis logic banana
- ML prediction integrate karna
- Code architecture decide karna
- GitHub repo manage karna
- Weekly progress sync lena

Deliverables:

- `risk_engine/`
- `ml/`
- Integration testing

Tum “core intelligence” sambhaloge.

2 Data Engineer – Data Pipeline Owner

Ye banda system ka foundation banayega.

Responsibilities:

- Dataset collection
- Cleaning (missing values, outliers)
- Feature engineering

- Normalization
- Database me structured insertion
- Basic EDA graphs

Deliverables:

- `data/loader.py`
- `data/preprocessor.py`
- Clean dataset ready for risk engine

Agar data ganda hua → pura project hil jayega. Is role ko lightly mat lena.

3 Backend Developer – Flask + API Owner

Iska kaam ML ko web se connect karna.

Responsibilities:

- Flask app structure
- API routes:
 - `/dashboard`
 - `/zone/<id>`
 - `/api/risk`
 - `/admin/upload`
- JSON responses
- DB connection handling
- Admin authentication (basic)

Deliverables:

- `app.py`
- API endpoints working
- DB models integration

Ye banda “system glue” hai.

4 Frontend Developer – UI + Visualization Owner

Judges ko ye banda impress karega.

Responsibilities:

- Dashboard layout
- Zone selection UI
- Risk color coding
- Chart.js graphs (monthly trend)
- Clean CSS design

Deliverables:

- `templates/`
- `static/css/`
- `static/js/`

Focus: clarity, not fancy animation.

5 (If Available) Documentation + Research Lead

Agar 5th member hai:

Responsibilities:

- Problem research
- Related work summary
- PPT slides
- Architecture diagram
- Feature explanation
- Testing & validation report

SIH me ye banda gold hota hai.

Agar 4 log hi hain:

Ye role tumhare saath merge hoga.

Work Phase Plan (Important)

Phase 1 (Week 1–2)

- Dataset finalize
- DB schema design
- Basic Flask setup

Phase 2 (Week 3–4)

- Risk calculation working
- Data pipeline stable
- Basic dashboard visible

Phase 3 (Month 2)

- Seasonal analysis

- Prediction model
- Resource recommendation

Phase 4

- Polishing
- Testing
- Documentation

Leader Advice (Very Important)

Galti jo teams karti hain:

- ✗ Sabko ML de dena
- ✗ Sabko “thoda-thoda” karwana
- ✗ No weekly review

Tum karo:

- ✓ Har banda apna module own kare
- ✓ Har Sunday progress check
- ✓ Integration early start karo